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March 14, 2025

Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO)
National Science Foundation

Subject: AI Action Plan

Comments submitted via [REDACTED]

Dear Ms. Plimpton,

On behalf of the Society of Cardiovascular Computed Tomography (SCCT), we appreciate the opportunity to provide comments on the Request for Information (RFI) regarding the development of an AI Action Plan. SCCT is a recognized leader in cardiovascular computed tomography (CT) education, advocacy, and research, representing a diverse membership of physicians, scientists, and healthcare professionals dedicated to advancing the use of CT in cardiovascular care.

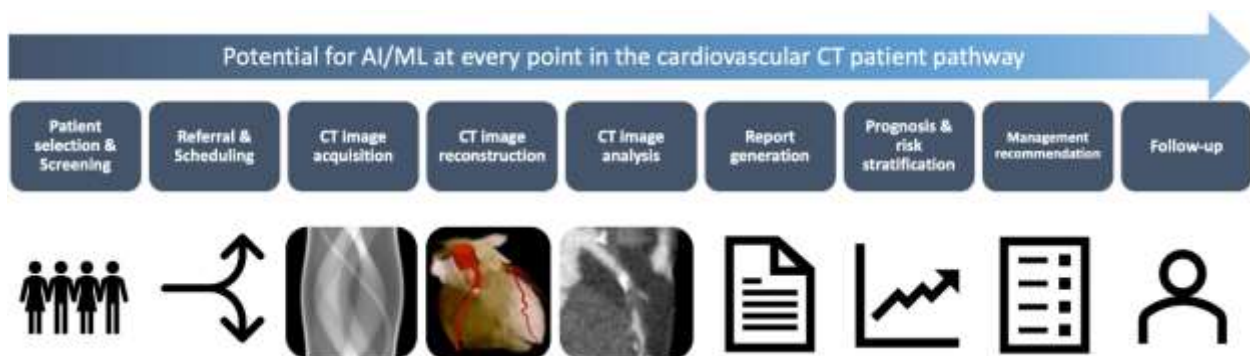
Artificial intelligence (AI) is integral to modern cardiovascular imaging, playing a crucial role in research, treatment, clinical decision-making, and healthcare delivery. The increasing adoption of AI across healthcare underscores the necessity for its inclusion in federal policy discussions to ensure responsible integration, innovation, and patient safety.

AI applications are also increasingly used across a number of patient care pathways:

- **Medical procedures:** Supports diagnostic image acquisition, reducing errors and improving workflow, resulting in lower medical costs and supporting patient safety.
- **Image interpretation:** Automates aspects of time-consuming tasks for clinicians and standardizes reporting of results resulting in higher accuracy, quality care and more time for the patient.
- **Clinical decision making:** Help assess risk prediction and resulting decision support, which leads to better patient/treatment management.

Cardiovascular CT is now a key investigation for patients with known or suspected cardiac disease; and is also increasingly adopted in primary prevention assessments for those populations at risk. Some reports suggest that implementing guideline recommendations will require in 7-10x more cardiovascular CT exams. Therefore, there is an enormous need for AI to assist and improve patient care at every point in the cardiovascular CT pathway.

Among other applications, AI-driven analyses provide detailed non-invasive assessments of the coronary arteries permitting optimal prevention strategies and treatment of patients at-risk of or with established coronary artery diseases (angina or prior heart attack). There is potential for AI and machine learning (AI/ML) to improve patient care at every point in the cardiovascular CT pathway.



Several aspects of the traditionally time-consuming cardiovascular CT analysis can now be automated by AI, freeing up physicians for human-dependent aspects of patient care; and providing them with a level of information to implement efficient, personalized care to every patient. For example, AI-based analyses can rapidly identify and quantify coronary calcification; improve the accuracy of coronary artery lumens segmentation; accurately quantify coronary stenoses and allow virtual stress assessments; and rapidly calculate total atherosclerotic plaque volumes, providing unprecedented risk assessment beyond standard human cognition.

Given the transformative impact of AI on cardiovascular CT, as well as on cardiovascular medicine as a whole, SCCT recently (August 2024) published a white paper [“Artificial Intelligence and Machine Learning for Cardiovascular Computed Tomography \(CCT\): A White Paper of the SCCT”](#) which summarises the current state-of-the-art advances, highlights challenges and considerations for AI implementation in both research and clinical settings, and explores the future of this technology.

In this context, SCCT urges the Administration to recognize the critical role of AI in cardiovascular medicine and include healthcare as a central focus in the AI Action Plan.

Key priorities for AI in healthcare policy

1. Reimbursement pathways for AI-driven innovation

- AI-driven tools enhance efficiency and patient outcomes, but reimbursement models must evolve to support their adoption.

- Federal policy should facilitate reimbursement mechanisms that encourage responsible AI-driven innovation in clinical practice.

2. Responsibility and liability for AI-related errors

- Clear regulatory frameworks are necessary to define liability in AI-assisted medical decision-making.
- Policies should establish accountability while ensuring AI remains a tool that supports, rather than replaces, medical professionals.

3. Assuring data robustness, safety and effectiveness

- AI applications in healthcare rely on high-quality, diverse, and representative datasets.
- Policymakers must ensure AI algorithms meet rigorous standards for validation and trustworthiness to maintain trust and efficacy.

4. AI as a supportive tool, not a replacement for clinical judgment

- AI should complement rather than dictate clinical decisions, particularly in prior authorization and diagnostic interpretation.
- Regulatory frameworks must reinforce AI's role in augmenting healthcare professionals rather than supplanting their expertise.

As the Administration develops its AI Action Plan, it is imperative that healthcare be included as a priority sector. SCCT welcomes the opportunity to collaborate with policymakers to shape an AI framework that fosters innovation while safeguarding patient care and clinical integrity. We appreciate your consideration of these critical issues and look forward to engaging further in this important policy discussion. Please feel free to contact us for further dialogue.

Sincerely,

Ahmad Slim, MD, FSCCT
Chair, SCCT Health Policy & Practice Committee